

Colin Keith

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SUMMARY

Entry-level data scientist with strong programming skills in Python, R and SQL and hands-on experience applying regression, classification, decision trees and neural networks to time series and statistical analysis. Built and fine-tuned predictive models through academic research projects and teaching assistant roles. Proven ability to communicate complex methodologies clearly to diverse audiences and collaborate with cross-functional teams. Seeking to leverage theoretical knowledge and technical expertise to develop and optimize AI-driven solutions.

PROFESSIONAL EXPERIENCE

EY

Jan 2022 - Apr 2023

Assurance Staff

Saint John, NB

- Performed audits for clients of various sizes across Canada and the Bahamas as part of a professional assurance team.
- Reviewed and analyzed critical client documents, including invoices, receipts, and bills of lading.
- Updated and maintained complex Excel workbooks across multiple years and diverse client profiles to support more efficient audit processes.
- Coordinated daily with teammates and clients to ensure audit accuracy and deadline compliance.

RBC

Dec 2020 - May 2021

Banking Advisor

Moncton, NB

- Served as the primary point of contact for customer concerns and questions at the Moncton advice centre.
- Resolved customer issues regarding bank accounts and various financial products.
- Provided tailored information and product recommendations based on individual customer needs by analyzing client data and account histories.
- Coordinated technical and financial solutions with advisors and specialized staff.

Rocmaura Nursing Home

Jun 2014 - Aug 2019

Environmental Supervisor

Saint John, NB

- Managed the cleaning and maintenance of public spaces and private resident rooms.
- Ensured the regular and secure handling of hazardous materials and equipment.
- Applied time management and team coordination skills daily to meet facility standards, leveraging complex problem-solving to handle unexpected maintenance challenges.

Dalhousie University & University of New Brunswick

Sep 2015 - Apr 2020

Teaching and Research Assistant

Various Locations

- Summarized academic articles regarding interest rate convergence across regional time series for research purposes using R and Python.
- Responsible for marking assignments and tests for undergraduate courses.
- Prepared and delivered lectures and labs on entry-level R programming for undergraduate students.
- Provided tutoring and oversaw exams to support student success.

EDUCATION

New Brunswick Community College

Sep 2024 - Apr 2026

Information Technology: Business and Advanced Analytics

Saint John, NB

- **GPA:** 3.5

Dalhousie University

Apr 2020

Master of Arts, Economics

Halifax, NS

- **GPA:** 3.3

University of New Brunswick

Apr 2019

Bachelor of Business Administration, Honours in Economics

Fredericton, NB

- **GPA:** 3.6
- **Achievements:** First Class Honours, Tutored at the Department of Economics help desk

PROJECTS

Capstone Project: Price Elasticity Modeling

Jan 2026 - Apr 2026

Moosehead Breweries & NBCC

- Extracted and processed over 16 million sales and 40 million inventory records from SQL databases, constructing an automated Python data pipeline to support reproducible analysis

- Built a neural network price elasticity model (MLP Regressor) using a log-log econometric framework, achieving an R^2 of ~ 0.80 ; estimated an average price elasticity of demand of -0.23 , confirming inelastic consumer behaviour consistent with Canadian alcohol industry benchmarks.
- Benchmarked the neural network against Linear Regression, Random Forest, and XGBoost models; elasticity findings were consistent across all model families, strengthening confidence in the results.
- Produced a seasonal price elasticity heatmap identifying weekday winter periods as the highest-impact windows for promotions, delivering an actionable pricing strategy roadmap to Moosehead.
- Built a companion Random Forest stockout risk classifier (ROC-AUC: 0.91, F1: 0.71) to predict low-inventory events, enabling proactive replenishment decisions and reducing missed sales opportunities.
- Engineered features such as cyclical time encoding, log transformations, one-hot encoding, and SelectKBest selection within a scalable Python data pipeline, reducing memory footprint from 1,855 MB to 148 MB
- Developed the end-to-end analysis in Python using pandas, NumPy, scikit-learn, XGBoost, and matplotlib/seaborn; version-controlled the Quarto project repository to ensure fully reproducible results

SKILLS

- **Technical Skills:** R, Python, SQL, Excel, PowerBI, Data Pipelines, Machine Learning Libraries
- **Machine Learning:** Linear Regression, Logistical Classification, Decision Trees, Random Forest, Lasso/Ridge Regression, Neural Networks, Time Series Analysis
- **Professional Skills:** Written and Verbal Communication, Teaching and Presentation, Team Coordination, Conflict Resolution, Complex Problem-Solving